Adelaide University

COMP6001 Computer Vision and Multimodal Machine Learning

Investigating the Impact of Classical Image Deblurring on Cross-Domain Object Detection Performance

Assignment Report

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Abstract

Motion blur is a common problem in computer vision because it reduces image clarity and makes it harder for object detection models to recognise important features. This project investigates whether classical image restoration methods can improve object detection on blurred images.

Blurred images from the GoPro dataset were restored using Wiener filtering and Richardson–Lucy deconvolution. Their quality was measured using PSNR, SSIM, Laplacian variance and runtime. A pretrained YOLOv8n model was then used to compare object detection performance across blurred, restored and sharp image domains.

The results showed that Richardson–Lucy restoration produced better visual quality than Wiener filtering. However, even though the restored images looked clearer, they did not improve object detection performance. Models trained on sharp and blurred images achieved better precision, recall and mAP values than models trained on restored images.

Overall, the findings suggest that visually sharper images do not always provide better features for object detection. In this project, models trained on sharp images gave the most reliable performance across different image domains.

1 Introduction

Motion blur is often caused by camera shake, fast-moving objects, low lighting conditions or long exposure times. This type of blur can reduce image quality and make it difficult to see edges, textures and object boundaries clearly. Since object detection models rely on these features, blurred images can reduce detection accuracy.

Image restoration methods are often used to improve blurred images before they are used for computer vision tasks. Classical methods such as Wiener filtering and Richardson–Lucy deconvolution are still widely used because they are simple, fast and do not need large amounts of training data (Gonzalez and Woods 2018). However, even if these methods improve how an image looks, it does not necessarily mean they will improve object detection performance.

This project compares blurred, restored and sharp images using YOLOv8 models. The aim is to understand whether classical deblurring methods can help object detection and whether restored images are useful for training.

2 Research Questions

RQ1. Can classical image restoration methods improve object detection performance on blurred images?

RQ2. Which restoration method provides the best balance between image quality, runtime and downstream object detection performance?

RQ3. How do models trained on sharp, blurred and restored images perform across different image domains?

3 Datasets and Data Preparation

3.1 GoPro Deblur Dataset

The GoPro Deblur dataset contains paired blurred and sharp images captured in realistic motion scenarios (Kupyn et al. 2018). It was used to evaluate image restoration quality because it provides ground-truth sharp images for comparison.

The dataset used in this project contained 1,029 paired images. These images were grouped into low, medium and high blur categories using a sharpness-based blur estimate.

Table 1: GoPro restoration dataset summary

Dataset	Total Images	Low Blur	Medium Blur	High Blur
GoPro Deblur Dataset	1,029	343	343	343

3.2 COCO and CocoBlur Dataset

The object detection stage used COCO 2017 validation images and their corresponding blurred and restored versions (Lin et al. 2014). Each image existed in three domains:

- Sharp
- Blurred
- Richardson–Lucy restored

The COCO validation pairing stage contained 5,000 matched images.

Table 2: COCO dataset summary

Dataset	Total Images	Low Blur	Medium Blur	High Blur
COCO Validation Dataset	5,000	1,667	1,666	1,667

For the final YOLO training stage, 197 images were selected for each image domain. The dataset contained an average of approximately 6.6 objects per image.

Table 3: Final YOLO training subset

Domain	Number of Images
Sharp	197
Blurred	197
Richardson–Lucy	197

3.3 Train, Validation and Test Preparation

To support reproducibility and fair comparison, the final subset was organised into train, validation and test folders for each domain. The same annotations were used across all domains because each version corresponded to the same underlying scene content. This ensured that comparisons focused on image quality differences rather than label inconsistencies.

4 Methodology

4.1 Restoration Methods

Two classical restoration methods were evaluated.

4.1.1 Wiener Filtering

Wiener filtering is a frequency-domain restoration method that attempts to reverse blur while reducing noise amplification (Wiener 1949). The Wiener equation is:

$$\hat{F}(u,v) = \left(\frac{H^*(u,v)}{|H(u,v)|^2 + K} \right) G(u,v)$$

where $\hat{F}(u,v)$ is the restored image, $H(u,v)$ is the blur kernel, $H^*(u,v)$ is the complex conjugate of the kernel, $G(u,v)$ is the blurred image, and K is a stabilisation constant.

4.1.2 Richardson–Lucy Deconvolution

Richardson–Lucy deconvolution is an iterative image restoration technique that refines an image estimate using a blur kernel (Richardson 1972, Lucy 1974). The update rule is:

$$f_{k+1}(x,y) = f_k(x,y) \left[\frac{g(x,y)}{f_k(x,y) * h(x,y)} * h(-x,-y) \right]$$

where $f_k(x,y)$ is the current sharp estimate, $g(x,y)$ is the blurred image, $h(x,y)$ is the blur kernel, and $*$ denotes convolution.

4.2 Image Quality Metrics

The restoration methods were evaluated using PSNR, SSIM, Laplacian variance and runtime.

PSNR was calculated as:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

SSIM was calculated as:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

PSNR measures pixel-level similarity, while SSIM measures structural similarity and visual consistency (Wang et al. 2004).

4.3 Object Detection Evaluation

A pretrained YOLOv8n model was first used to compare sharp, blurred and restored images (Ultralytics 2024). Mean detection count, confidence score and runtime were measured.

Separate YOLOv8 models were then trained on each image domain and evaluated using:

- precision
- recall
- mAP@50
- mAP@50–95

5 Experimental Setup

The restoration stage was evaluated on 300 GoPro image pairs. Images were resized before processing to improve runtime efficiency. The object detection stage used YOLOv8n because it provides a practical balance between accuracy and computation for a small-to-medium scale experiment.

The final training subset contained 197 images per domain. The same annotations were retained across sharp, blurred and restored versions so that the effect of image quality could be studied directly.

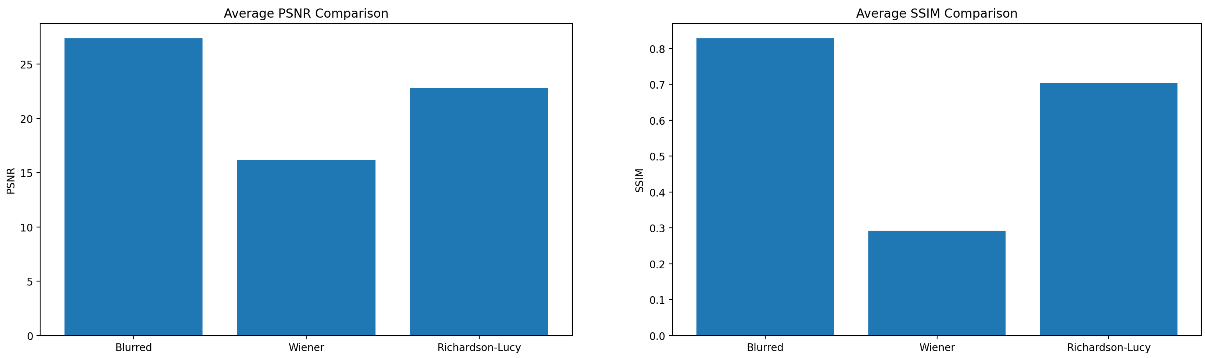
6 Results

6.1 Restoration Performance

Richardson–Lucy restoration achieved stronger PSNR and SSIM values than Wiener filtering. Although Wiener filtering produced the highest Laplacian variance, this result was mainly caused by amplified noise and ringing artifacts rather than genuine sharpness.

Table 4: Average restoration performance

Method	PSNR	SSIM	Laplacian Variance	Runtime (s)
Blurred	27.38	0.828	246.53	0.000
Wiener	16.17	0.291	1612.58	0.193
Richardson–Lucy	22.80	0.704	345.83	0.297



(a) Average PSNR comparison

(b) Average SSIM comparison

Figure 1: Restoration quality comparison across image domains.

6.2 Pretrained YOLO Performance

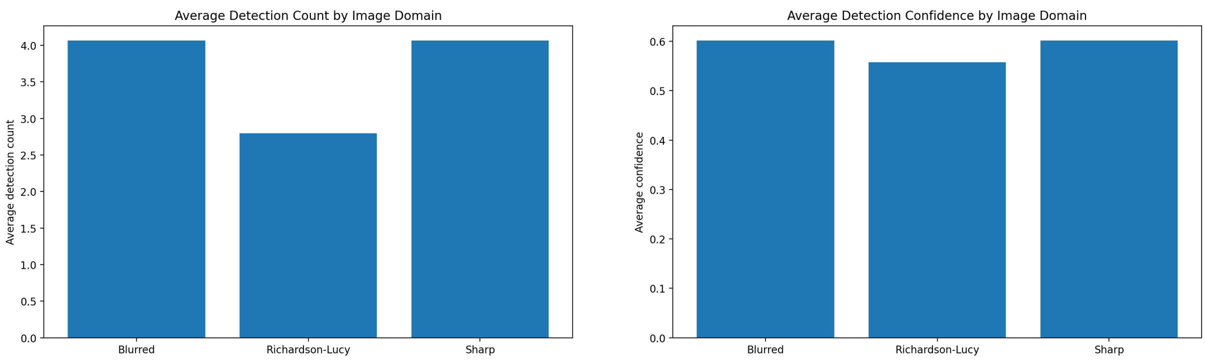
The pretrained YOLO model performed almost identically on sharp and blurred images. Richardson–Lucy images produced fewer detections and lower confidence values. This suggests that restoration artifacts weakened the local features used by the detector.

Table 5: Pretrained YOLOv8 detection comparison

Domain	**Detection Count**	**Confidence**	**Runtime (s)**
Blurred	4.065	0.601	0.176
Richardson–Lucy	2.795	0.558	0.105
Sharp	4.065	0.601	0.111

The percentage drop in detection count from blurred to restored images was:

$$\frac{4.065 - 2.795}{4.065} \times 100 \approx 31.2\%$$



(a) Detection count

(b) Confidence

Figure 2: Pretrained YOLOv8 comparison across blurred, restored and sharp image domains.

6.3 Final YOLO Training Results

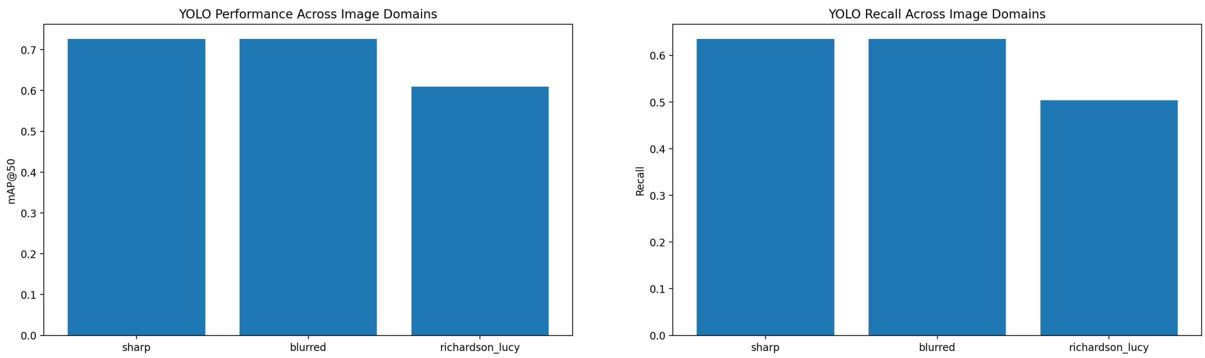
The final YOLO models trained on sharp and blurred images achieved identical performance, while the model trained on restored images performed worse across all metrics.

Table 6: Final YOLO cross-domain comparison

Domain	**Precision**	**Recall**	**mAP@50**	**mAP@50–95**
Sharp	0.797	0.635	0.726	0.562
Blurred	0.797	0.635	0.726	0.562
Richardson–Lucy	0.743	0.504	0.609	0.451

The mAP@50 drop from the sharp model to the restored-image model was:

$$\frac{0.726 - 0.609}{0.726} \times 100 \approx 16.1\%$$



(a) mAP@50 comparison **(b)** Recall comparison

Figure 3: Final detection performance across image domains.

7 Discussion

The results showed that better image quality does not always lead to better object detection performance. Richardson–Lucy restoration produced higher PSNR and SSIM values than Wiener filtering, which means the restored images were visually closer to the original sharp images. However, the restored images still gave lower detection accuracy.

One reason for this may be that the restoration process introduced ringing effects and unstable edges around objects. These changes may not look very noticeable to a person, but they can affect the features that YOLO uses to detect objects.

Another important finding was that sharp and blurred images gave very similar detection performance. This suggests that moderate motion blur may not affect YOLOv8 as much as expected.

8 Failure Cases and Critical Analysis

Some restored images looked sharper to the human eye but still performed worse during detection. In several examples, Richardson–Lucy restoration created extra edges around objects or distorted small textures. This sometimes reduced the confidence score or caused objects to be missed completely.

Wiener filtering was faster than Richardson–Lucy, but it often created strong artifacts and unrealistic sharpness. Richardson–Lucy gave better visual quality overall, but it still did not improve detection accuracy.

This shows that the best-looking restored image is not always the best image for a detection model.

9 Responsible AI Use, Reproducibility and Ethics

AI tools were used during the project mainly for troubleshooting code, checking package compatibility, improving notebook organisation and refining the report structure. However, all code, results and explanations were tested and reviewed manually before being included.

The project was designed to be reproducible by keeping the dataset preparation process consistent, saving intermediate outputs and clearly recording the evaluation results. Public datasets were used throughout the project, so there were no privacy or ethical concerns related to personal data.

10 Limitations and Future Work

This project had several limitations:

- only two classical restoration methods were tested
- only one YOLO architecture was used
- the final training subset was relatively small
- simplified blur kernels were used
- runtime limitations restricted restoration tuning

Future work could investigate deep-learning-based deblurring methods, larger datasets and more advanced detection architectures.

11 Conclusion

This project investigated whether classical image restoration methods can improve object detection on blurred images. Richardson–Lucy deconvolution produced better image quality than Wiener filtering, but this did not improve detection accuracy.

The final results showed that models trained on sharp images performed best overall, while restored-image models were more affected by artifacts introduced during restoration.

Overall, the project shows that improving the visual quality of an image does not always improve the quality of the features used by an object detection model.

References

Gonzalez, R.C. and Woods, R.E. 2018, *Digital Image Processing*, 4th edn, Pearson, New York.

Kupyn, O., Budzan, V., Mykhailych, M., Mishkin, D. and Matas, J. 2018, 'DeblurGAN: Blind Motion Deblurring Using Conditional Adversarial Networks', *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8183–8192.

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P. and Zitnick, C.L. 2014, 'Microsoft COCO: Common Objects in Context', *European Conference on Computer Vision*, pp. 740–755.

Lucy, L.B. 1974, 'An Iterative Technique for the Rectification of Observed Distributions', *The Astronomical Journal*, vol. 79, no. 6, pp. 745–754.

Richardson, W.H. 1972, 'Bayesian-Based Iterative Method of Image Restoration', *Journal of the Optical Society of America*, vol. 62, no. 1, pp. 55–59.

Ultralytics 2024, *YOLOv8 Documentation*, viewed 1 April 2026, ¡https://docs.ultralytics.com¿.

Wang, Z., Bovik, A.C., Sheikh, H.R. and Simoncelli, E.P. 2004, 'Image Quality Assessment: From Error Visibility to Structural Similarity', *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612.

Wiener, N. 1949, *Extrapolation, Interpolation and Smoothing of Stationary Time Series*, MIT Press, Cambridge.